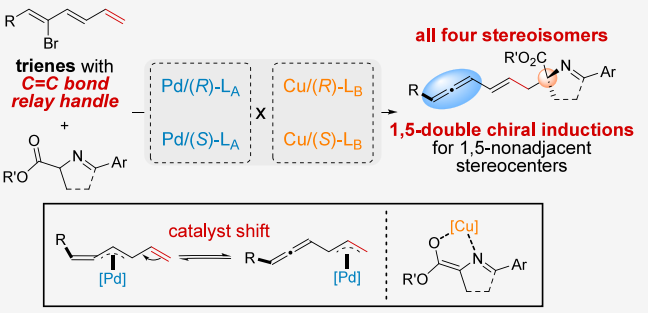
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Synergistic Pd/Cu-Catalyzed 1,5-Double Chiral Inductions

Jiacheng Zhang, Yicong Luo, En Zheng, Xiaohong Huo, Shengming Ma,* and Wanbin Zhang*

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ABSTRACT: Much attention has been focused on the catalytic asymmetric creation of single chiral centers or two adjacent stereocenters. However, the asymmetric construction of two nonadjacent stereocenters is of significant importance but is challenging because of the lack of remote chiral induction models. Herein, based on a C═C bond relay strategy, we report a synergistic Pd/Cu-catalyzed 1,5-double chiral induction model. All four stereoisomers of the target products bearing 1,5-nonadjacent stereocenters involving both allenyl axial and central chirality could be obtained divergently by simply changing the combination of two chiral catalysts with different configurations. Control experiments and DFT calculations reveal a novel mechanism involving 1,5-oxidative addition, contra-thermodynamic η^3-allyl palladium shift, and conjugate nucleophilic substitution, which play crucial roles in the control of reactivity, regio-, enantio-, and diastereoselectivity. It is expected that this C═C bond relay strategy may provide a general protocol for the asymmetric synthesis of structural motifs bearing two distant stereocenters.



INTRODUCTION

Asymmetric catalysis is one of the most efficient protocols for creating optically active compounds, which is of fundamental importance in life science, medicinal chemistry, and materials science.[1] A large number of synthetic approaches have been developed to achieve the asymmetric construction of single stereocenters and the simultaneous construction of two adjacent stereocenters (Figure 1). However, very limited

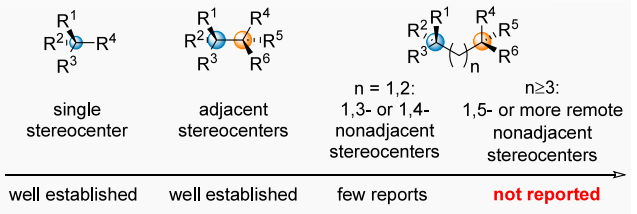
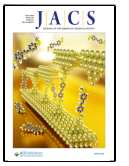


Figure 1. Asymmetric catalysis targets.

reports have been disclosed on the one-step synthesis of structural motifs bearing nonadjacent stereocenters.[2−5] It is expected that the challenge will increase dramatically as the distance between two stereogenic centers increases due to the lack of reliable chiral induction models for both chiral entities. To date, no example of the asymmetric catalytic construction of 1,5-nonadjacent or more remote stereocenters has been reported. Over the past decade, the synergistic catalysis strategy has been proven to be a powerful synthetic tool to asymmetrically access structural motifs bearing two stereocenters in a single step, such as two adjacent,[6−8] 1,3-,[3] and 1,4-nonadjacent stereocenters[5] (Scheme 1A, top). Here, by leveraging the advantages of synergistic catalysis,[9] we envisioned that a relay strategy involving the shift of chiral catalysts could provide a viable solution to the challenging problems associated with the double chiral inductions for two nonadjacent stereocenters (Scheme 1A, bottom).

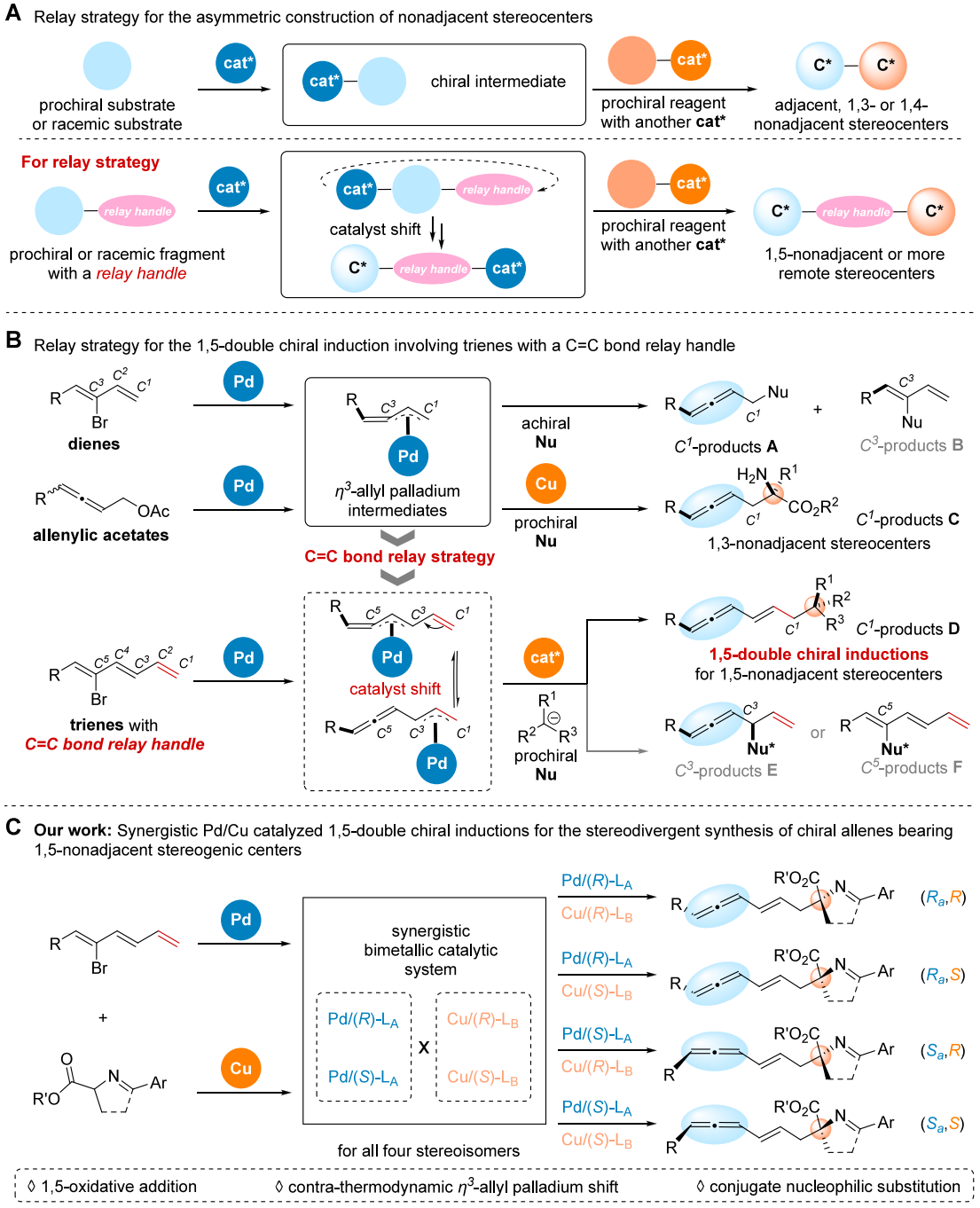
Allene chemistry has become an important branch in organic chemistry, pharmacy, and materials science due to their unique structural feature.[10] Consequently, synthetic organic chemists have been pursuing general and efficient methods for the catalytic asymmetric synthesis of chiral allenes.[11,12] The Hayashi group and Ogasawara−Takahashi group disclosed a Pd-catalyzed allenylation of nucleophiles with 3-bromo-1,3-dienes, demonstrating a remarkable regioselectivity for the C^1-position (**A**) over the C^3-position (**B**) or the construction of allenes with single axial chirality (Scheme 1B, top).[13] In our previous work, the structural motifs bearing 1,3-nonadjacent stereocenters (**C**) have been successfully obtained via a synergistic bimetallic catalytic system with racemic allenylic acetates and prochiral nucleophiles (Scheme 1B, middle).[3b] It is envisioned that a relay strategy would provide a protocol for 1,5-double chiral induction with 5-bromo-1,3,5-trienes containing a C═C bond acting as a relay handle, generating both

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Scheme 1. Asymmetric Catalysis of Compounds with Nonadjacent Stereocenters



axial and central chirality (Scheme 1B, bottom). However, for this strategy, a notable challenge is the asymmetric synthesis of the desired C^1-products (**D**) bearing 1,5-nonadjacent stereocenters with high enantio- and diastereoselectivities, as well as high regioselectivity among C^1-, C^3-, and C^5-products (**D**, **E**, **F**). Herein, based on the C=C bond relay strategy, we report a synergistic Pd/Cu-catalyzed 1,5-double chiral induction model for the first asymmetric construction of 1,5-nonadjacent stereocenters bearing both allenyl axial and central chirality with good stereoselectivity and excellent regioselectivity (Scheme 1C). Notably, all four possible stereoisomers could be synthesized in a stereodivergent manner by simply changing the combination of two chiral catalysts with different configurations.[14] Furthermore, control experiments and DFT calculations reveal a novel mechanism involving three key elementary reactions: 1,5-oxidative addition, contra-thermodynamic η^3-allyl palladium shift, and conjugate nucleophilic substitution, which clearly illustrate the importance of the C=C bond relay strategy in controlling reactivity and regio-, enantio-, and diastereoselectivity. It is expected that this relay strategy could provide a general platform for the asymmetric construction of structural motifs bearing two distant stereocenters and could change the manner in which chemists think about designing and building chiral molecules.[15]

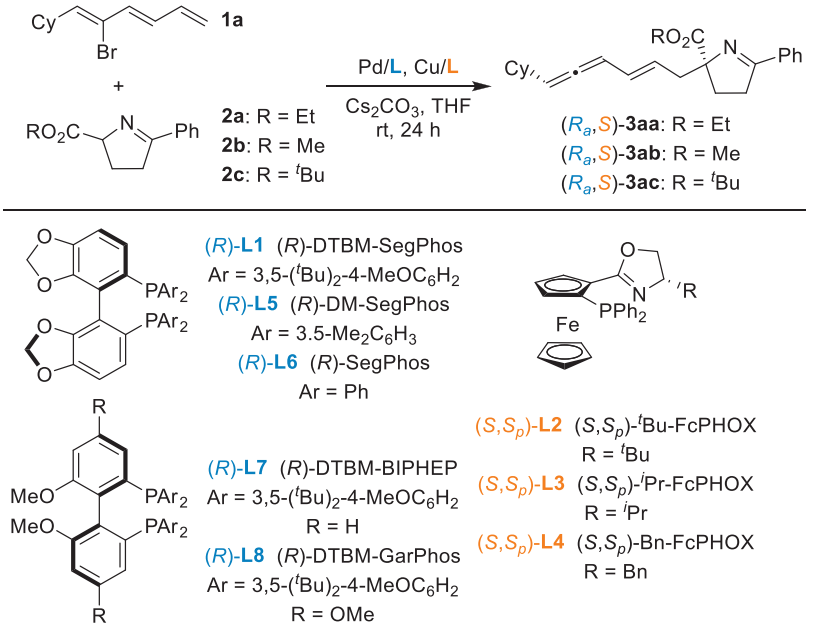
RESULTS AND DISCUSSION

To verify the feasibility of our hypothesis, cyclohexyl-substituted (3*E*,5*Z*)-5-bromo-1,3,5-triene **1a** was selected as

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Table 1. Optimization of the Reaction Conditions[a]



Entry	2	1:2	L (Pd)	L (Cu)	Yield (%)[b]	Ee (%)[c]	Dr[c]
1	**2a**	1:1.5	(*R*)-**L1**	(*S*,*S*$_p$)-**L2**	36	94/47	2:1
2	**2a**	1.5:1	(*R*)-**L1**	(*S*,*S*$_p$)-**L2**	90	>99	11:1
3	**2a**	1.5:1	(*R*)-**L1**	(*S*,*S*$_p$)-**L3**	92	>99	13:1
4	**2a**	1.5:1	(*R*)-**L1**	(*S*,*S*$_p$)-**L4**	36	98	6:1
5	**2a**	1.5:1	(*R*)-**L5**	(*S*,*S*$_p$)-**L3**	60	>99/76	3:1
6	**2a**	1.5:1	(*R*)-**L6**	(*S*,*S*$_p$)-**L3**	84	94/56	2:1
7	**2a**	1.5:1	(*R*)-**L7**	(*S*,*S*$_p$)-**L3**	75	98	13:1
8	**2a**	1.5:1	(*R*)-**L8**	(*S*,*S*$_p$)-**L3**	27	98	16:1
9	**2b**	1.5:1	(*R*)-**L1**	(*S*,*S*$_p$)-**L3**	77	>99	6:1
10	**2c**	1.5:1	(*R*)-**L1**	(*S*,*S*$_p$)-**L3**	46	>99	6:1
11[d]	**2a**	1.5:1	--	(*S*,*S*$_p$)-**L3**	0	ND	ND
12[e]	**2a**	1.5:1	(*R*)-**L1**	--	Trace	ND	ND

[a]Reaction conditions: **1a** (0.1−0.15 mmol, 1.0−1.5 equiv), **2a**−**2c** (0.10−0.15 mmol, 1.0−1.5 equiv), [Pd(η^3-allyl)Cl]$_2$ (2.5 mol %), (*R*)-**L1** or (*R*)-**L5**−**L8** (5.5 mol %), [Cu(CH$_3$CN)$_4$]PF$_6$ (5.0 mol %), (*S*,*S*$_p$)-**L2**−**L4** (5.5 mol %), Cs$_2$CO$_3$ (0.12 mmol, 1.2 equiv), and THF (anhydrous, 1.0 mL), rt, 24 h. [b]Isolated yield of all stereoisomers. [c]Determined by HPLC analysis. ND = not detected. [d]Without Pd catalyst. [e]Without Cu catalyst.

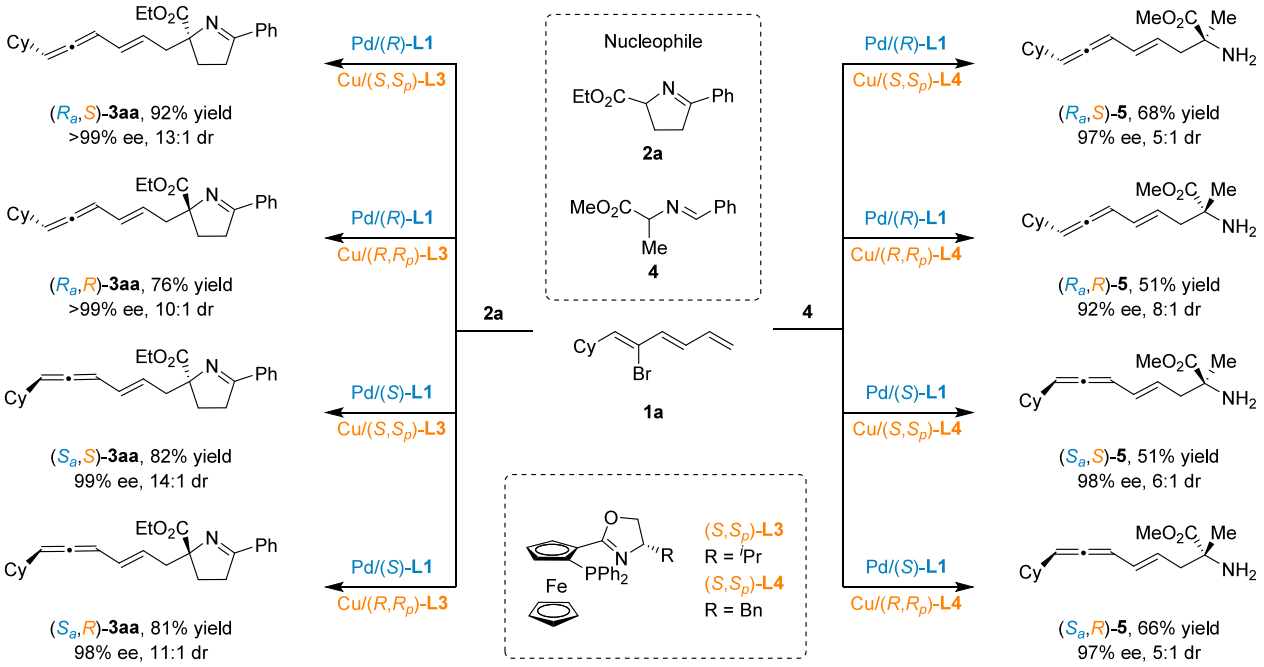
the model substrate for the asymmetric construction of 1,5-nonadjacent stereocenters by using imino ethyl ester **2a** as the nucleophile precursor and Cs$_2$CO$_3$ as the base with a [Pd/(*R*)-DTBM-SegPhos **L1** + Cu/(*S*,*S*$_p$)-^{t}Bu-FcPHOX **L2**] catalyst combination in THF. The desired product (*R*$_a$,*S*)-**3aa** was successfully obtained in 36% yield with 94%/47% ee and 2:1 dr without any regioisomers (>20:1 rr, Table 1, Entry 1). To our delight, the yield and diastereoselectivity were significantly improved to 90% yield and 11:1 dr by reversing the substrate ratio of triene **1a** and imino ester **2a** from 1:1.5 to 1.5:1 (Entry 2). Further screening of the ferrocene ligands (*S*,*S*$_p$)-^{i}Pr-FcPHOX **L3** and (*S*,*S*$_p$)-Bn-FcPHOX **L4** suggested that (*S*,*S*$_p$)-**L3** was the optimal ligand for the Cu catalyst, affording the desired product (*R*$_a$,*S*)-**3aa** in 92% yield with >99% ee and 13:1 dr (Entries 3 and 4). Then, a series of C_2-symmetric chiral bisphosphine ligands for the Pd catalyst [(*R*)-DM-SegPhos **L5**, (*R*)-SegPhos **L6**, (*R*)-DTBM-BIPHEP **L7**, and (*R*)-DTBM-GarPhos **L8**] were explored (Entries 5−8). These results revealed that a ligand with bulky substituents is necessary for efficient asymmetric inductions of two remote stereocenters,[16] and (*R*)-DTBM-SegPhos **L1** was the optimal ligand for the Pd catalyst. Subsequently, reactions with methyl- and *tert*-butyl-substituted imino esters **2b**−**c** were also conducted (Entries 9 and 10). No improvement in yield and diastereoselectivity was found. After determining the optimal conditions (see Entry 3), control experiments were conducted to clarify the specific role of the two chiral catalysts in this 1,5-double chiral induction model (Entries 11 and 12). No conversion was observed in the absence of any of the two chiral catalysts, verifying the necessity of the synergistic Pd/Cu catalytic system.

Encouraged by the successful optimization of our 1,5-double chiral induction model, we sought to further investigate the full

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Scheme 2. Synthesis of All Four Stereoisomers of 3aa and 5[a]



[a]Reaction conditions: **1a** (0.15 mmol, 1.5 equiv), **2a** or **4** (0.10 mmol, 1.0 equiv), [(*R*)/(*S*)-**L1**-Pd]BF_4 (5.0 mol %), [Cu$(CH_3CN)_4$]PF_6 (5.0 mol %), (R,R_p)/(S,S_p)-**L3** or **L4** (5.5 mol %), Cs_2CO_3 (0.12 mmol, 1.2 equiv), THF (anhydrous, 1.0 mL), rt, 24 h. For products **5**, hydrolysis with citric acid is performed to remove the protecting group of the amine group. The regioisomeric ratios of all above reactions are >20:1.

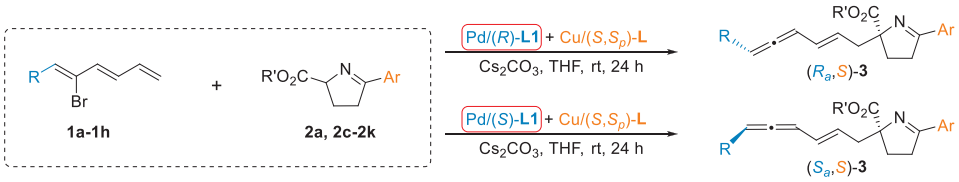
control of the absolute and relative configurations of both allenyl axial chirality and remote central chirality (Scheme 2). Indeed, by simply changing the ligand for the Cu catalyst from (S,S_p)-**L3** to (R,R_p)-**L3** while keeping the same ligand (*R*)-**L1** for the Pd catalyst, the complementary diastereoisomer (R_a,*R*)-**3aa** was successfully obtained in high yield with excellent enantio- and diastereoselectivity (76% yield, >99% ee, and 10:1 dr). Then, the remaining two stereoisomers [(S_a,*S*)-**3aa** and (S_a,*R*)-**3aa**] were also successfully prepared with good yields and excellent stereoselectivities using [Pd/(*S*)-**L1** + Cu/(S,S_p)-**L3**] and [Pd/(*S*)-**L1** + Cu/(R,R_p)-**L3**] catalyst combinations (for (S_a,*S*)-**3aa**, 82% yield, 99% ee, and 14:1 dr; for (S_a,*R*)-**3aa**, 81% yield, 98% ee, and 11:1 dr). Furthermore, aldimine ester **4** was applied to this catalyst system. The desired products **5** bearing remote axially chiral allene and chiral amino ester moieties were smoothly obtained in an enantio- and diastereodivergent manner with [Pd/**L1** + Cu/**L4**] catalyst combinations (51% to 68% yield, 92% to 98% ee, and 5:1 to 8:1 dr).

With the optimized reaction conditions in hand, the substrate scopes of trienes and imino esters were explored for this 1,5-double induction model (Scheme 3). First, reactions with trienes bearing a cyclohexyl (**1a**), cyclopentyl (**1b**), and cyclobutyl group (**1c**) were conducted with a [Pd/(*R*)-**L1** + Cu/(S,S_p)-**L3**] catalyst combination. The corresponding products (R_a,*S*)-**3aa**−**3ca** were smoothly obtained in moderate to high yields with satisfactory enantio- and diastereoselectivities [for (R_a,*S*)-**3aa**, 92% yield, >99% ee, and 13:1 dr; for (R_a,*S*)-**3ba**, 70% yield, >99% ee, and 9:1 dr; for (R_a,*S*)-**3ca**, 61% yield, 99% ee, and 3:1 dr]. These results suggested that the yield and diastereoselectivity highly correlate to the steric hindrance of the substituent groups of the trienes, probably due to the potential attractive dispersion interactions between the substrate and two chiral metal catalysts.[16] A similar phenomenon was also found in the asymmetric synthesis of (S_a,*S*)-**3aa**−**3ca** with a [Pd/(*S*)-**L1** + Cu/(S,S_p)-**L3**] catalyst combination. Furthermore, the triene **1d** bearing an *iso*-propyl group was successfully applied in this reaction, affording the desired products (R_a,*S*)-**3da** and (S_a,*S*)-**3da** in good yields and with excellent regio-, enantio-, and diastereoselectivities [for (R_a,*S*)-**3da**, 84% yield, >99% ee, and 10:1 dr; for (S_a,*S*)-**3da**, 61% yield, 98% ee, and 13:1 dr]. To our delight, the substrates containing 1°-alkyl groups (**1e**−**1g**) were also compatible with this catalyst system,[17] affording the corresponding products [(R_a,*S*)-**3ea**−**3ga** and (S_a,*S*)-**3ea**−**3ga**] in moderate yields and with good to excellent enantio- and diastereoselectivities (40% to 61% yield, 99% to >99% ee, and 6:1 to 14:1 dr). Finally, a triene bearing a sterically bulky ^{t}Bu group (**1h**) was investigated. The desired products (R_a,*S*)-**3ha** and (S_a,*S*)-**3ha** were obtained with good enantio- and diastereoselectivities but in low yields [for (R_a,*S*)-**3ha**, 41% yield, 96% ee, and 11:1 dr; for (S_a,*S*)-**3ha**, 48% yield, 92% ee, and 16:1 dr]. To evaluate the induction effect of a triene substrate bearing a chiral stereocenter, the substrate (*S*)-**1i** derived from (−)-citronellal was applied in this reaction using [Pd/(*R*)-**L1** + Cu/(S,S_p)-**L3**] and [Pd/(*S*)-**L1** + Cu/(S,S_p)-**L3**] catalyst combinations. The desired products [(S,R_a,*S*)-**3ia** and (S,S_a,*S*)-**3ia**] were successfully obtained in 56% and 57% yield and 9:1 and 7:1 dr, respectively. No significant differences in yield and stereoselectivity were observed. These results suggested that the existing chiral motif has negligible asymmetric induction on the axial and central chirality. Subsequently, further condition optimization was carried out to improve the yield of the reaction with **1h** (Table S2 in Supporting Information). Fortunately, the yield of product (R_a,*S*)-**3hc** could be increased to 79% with >99% ee and >20:1 dr after adding a small amount of water with a **1h**/**2c**/[Pd/(*R*)-**L1**]/[Cu/(S,S_p)-**L2**] ratio of 1:1.5:0.05:0.05 (Scheme 3,

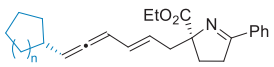
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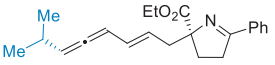
Scheme 3. Substrate Scope of Trienes and Imino Esters[a]



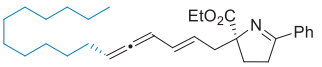
Condition **A**: Pd/(*R*)-**L1** + Cu/(*S*,S_p)-**L3**



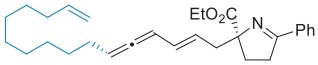
n = 2: (*R*$_a$,*S*)-**3aa**, 92% yield, >99% ee, 13:1 dr
n = 1: (*R*$_a$,*S*)-**3ba**, 70% yield, >99% ee, 9:1 dr
n = 0: (*R*$_a$,*S*)-**3ca**, 61% yield, 99% ee, 3:1 dr



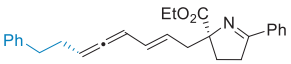
(*R*$_a$,*S*)-**3da**
84% yield
>99% ee, 10:1 dr



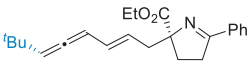
(*R*$_a$,*S*)-**3ea**
50% yield
>99% ee, 6:1 dr



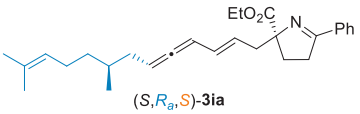
(*R*$_a$,*S*)-**3fa**
46% yield
99% ee, 14:1 dr



(*R*$_a$,*S*)-**3ga**
60% yield
>99% ee, 7:1 dr

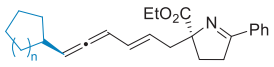


(*R*$_a$,*S*)-**3ha**
41% yield
96% ee, 11:1 dr

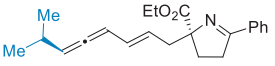


(*S*,*R*$_a$,*S*)-**3ia**
56% yield
9:1 dr

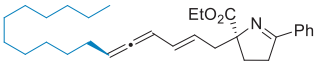
Condition **A**: Pd/(*S*)-**L1** + Cu/(*S*,S_p)-**L3**



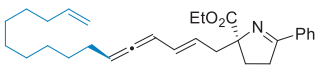
n = 2: (*S*$_a$,*S*)-**3aa**, 82% yield, 99% ee, 14:1 dr
n = 1: (*S*$_a$,*S*)-**3ba**, 58% yield, 99% ee, 10:1 dr
n = 0: (*S*$_a$,*S*)-**3ca**, 50% yield, 97% ee, 4:1 dr



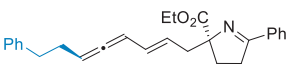
(*S*$_a$,*S*)-**3da**
61% yield
98% ee, 13:1 dr



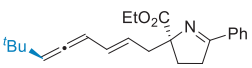
(*S*$_a$,*S*)-**3ea**
46% yield
>99% ee, 6:1 dr



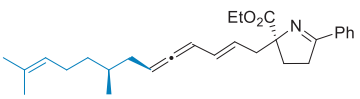
(*S*$_a$,*S*)-**3fa**
40% yield
>99% ee, 10:1 dr



(*S*$_a$,*S*)-**3ga**
61% yield
>99% ee, 7:1 dr

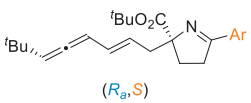


(*S*$_a$,*S*)-**3ha**
48% yield
92% ee, 16:1 dr

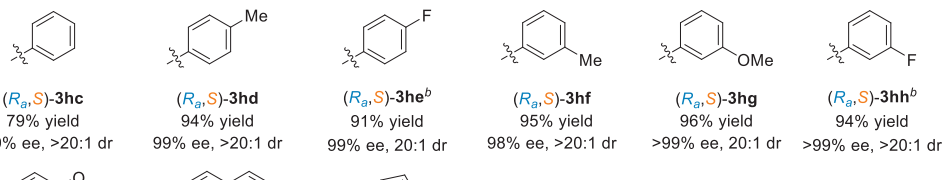


(*S*,*S*$_a$,*S*)-**3ia**
57% yield
7:1 dr

Condition **B**: Pd/(*R*)-**L1** + Cu/(*S*,S_p)-**L2**



(*R*$_a$,*S*)



(*R*$_a$,*S*)-**3hc**
79% yield
>99% ee, >20:1 dr

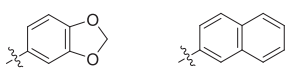
(*R*$_a$,*S*)-**3hd**
94% yield
99% ee, >20:1 dr

(*R*$_a$,*S*)-**3he**[b]
91% yield
99% ee, 20:1 dr

(*R*$_a$,*S*)-**3hf**
95% yield
98% ee, >20:1 dr

(*R*$_a$,*S*)-**3hg**
96% yield
>99% ee, 20:1 dr

(*R*$_a$,*S*)-**3hh**[b]
94% yield
>99% ee, >20:1 dr

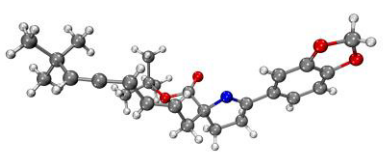


(*R*$_a$,*S*)-**3hi**
97% yield
>99% ee, >20:1 dr

(*R*$_a$,*S*)-**3hj**
95% yield
98% ee, >20:1 dr

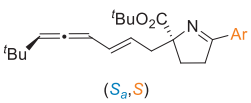


(*R*$_a$,*S*)-**3hk**
97% yield
99% ee, 18:1 dr

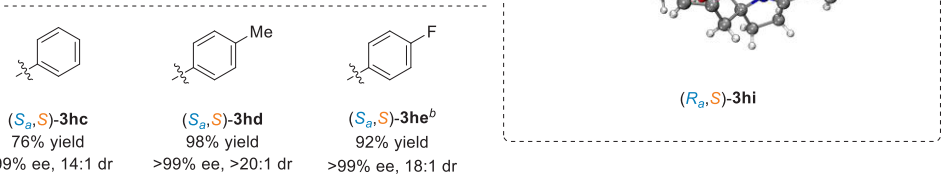


(*R*$_a$,*S*)-**3hi**

Condition **B**: Pd/(*S*)-**L1** + Cu/(*S*,S_p)-**L2**



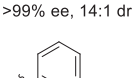
(*S*$_a$,*S*)



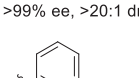
(*S*$_a$,*S*)-**3hc**
76% yield
>99% ee, 14:1 dr

(*S*$_a$,*S*)-**3hd**
98% yield
>99% ee, >20:1 dr

(*S*$_a$,*S*)-**3he**[b]
92% yield
>99% ee, 18:1 dr



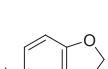
(*S*$_a$,*S*)-**3hf**
96% yield
99% ee, >20:1 dr



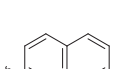
(*S*$_a$,*S*)-**3hg**
95% yield
>99% ee, >20:1 dr



(*S*$_a$,*S*)-**3hh**[b]
94% yield
98% ee, >20:1 dr



(*S*$_a$,*S*)-**3hi**
97% yield
>99% ee, 20:1 dr



(*S*$_a$,*S*)-**3hj**
95% yield
99% ee, >20:1 dr



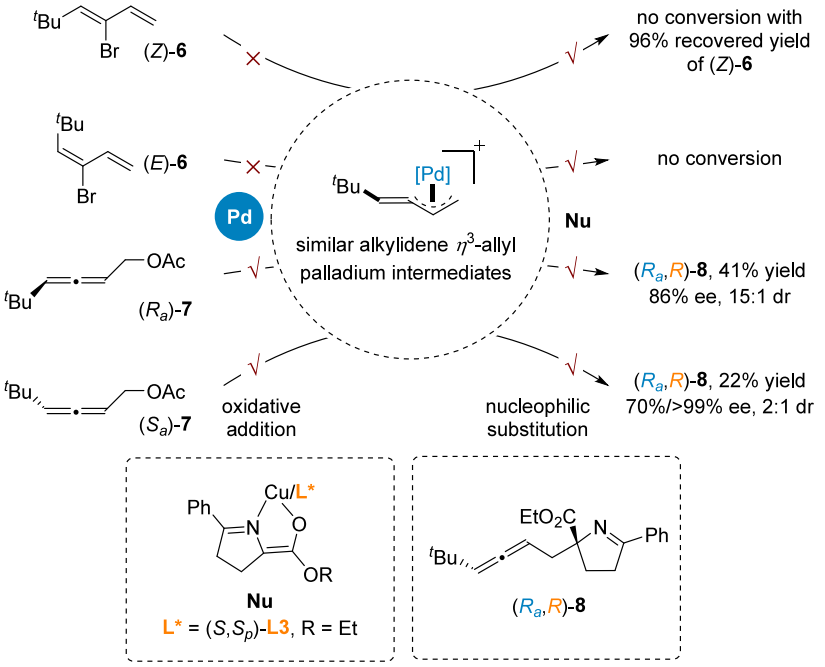
(*S*$_a$,*S*)-**3hk**
98% yield
95% ee, >20:1 dr

[a]Reaction conditions **A**: **1b**−**1h** (0.15 mmol, 1.5 equiv), **2a** (0.10 mmol, 1.0 equiv), [(*R*)/(*S*)-**L1**-Pd]BF_4 (5.0 mol %), [$Cu(CH_3CN)_4$]PF_6 (5.0 mol %), (*S*,S_p)-**L3** (5.5 mol %), Cs_2CO_3 (0.12 mmol, 1.2 equiv), THF (anhydrous, 1 mL), rt, 24 h. Reaction conditions **B**: **1h** (0.10 mmol, 1.0 equiv), **2c**−**2k** (0.15 mmol, 1.5 equiv), [(*R*)/(*S*)-**L1**-Pd]BF_4 (5.0 mol %), [$Cu(CH_3CN)_4$]PF_6 (5.0 mol %), (*S*,S_p)-**L2** (5.5 mol %), Cs_2CO_3 (0.12 mmol, 1.2 equiv), THF (anhydrous, 1.0 mL), H_2O (5 μL, 0.28 mmol, 2.8 equiv), rt, 24 h. The regioisomeric ratios of all above reactions are >20:1. [b]At 0 °C.

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Scheme 4. Control Experiments with Dienes (*Z*)-6 and (*E*)-6 and Allylic Acetates (R_a)-7 and (S_a)-7



Condition **B**). It should be noted that water might be beneficial for the dissolution of Cs_2CO_3, which could increase the efficiency of the deprotonation of the nucleophilic precursor **2c**.[18] The complementary diastereoisomer (S_a,*S*)-**3hc** could also be smoothly prepared in 76% yield with >99% ee and 14:1 dr by simply switching the configuration of the ligand of the Pd catalyst. Then, under the optimized reaction conditions, a series of imino esters were investigated for the generality of this catalytic system in a diastereodivergent manner. These results suggested that both electron-donating and electron-withdrawing groups at the *para*-/*meta*-position of the phenyl ring were well tolerated, and the corresponding products [(R_a,*S*)-**3hc**−**3hh** and (S_a,*S*)-**3hc**−**3hh**] were obtained in 91%−96% yields with 98% to >99% ee and 18:1 to >20:1 dr. Moreover, the reactions with imino esters bearing disubstituted aryl (**2i**), naphthyl (**2j**), and furyl (**2k**) groups also proceeded well with excellent yields and excellent stereoselectivities (95 to 98% yield, 95% to >99% ee and 18 to >20:1 dr). The absolute configuration of (R_a,*S*)-**3hi** was determined by X-ray crystallographic analysis (CCDC: 2216926).

Mechanistic Studies. To gain insights into the mechanism of this 1,5-double chiral induction model based on a C=C bond relay strategy, control experiments and DFT calculations were conducted. For the oxidative addition with trienes, there are two possible models involving either 1,3-oxidative addition[19] or 1,5-oxidative addition.[20] To determine the most feasible pathway, reactions with dienes (*Z*)-**6** and (*E*)-**6** and enantiopure allenylic acetates (R_a)-7 and (S_a)-7 were conducted for the study of the 1,3-oxidative addition process with a [Pd/(*R*)-**L1** + Cu/(S,S_p)-**L3**] catalyst combination by using imino ester **2a** as the nucleophile precursor (Scheme 4). As a result, no conversion was observed for dienes (*Z*)-**6** and (*E*)-**6**, while allenylic acetates (R_a)-7 and (S_a)-7 gave the product (R_a,*R*)-**8** in 41% and 22% yields, respectively. It was worth noting that substrate (*Z*)-**6** could be recovered in 96% yield, which suggested that diene **6** does not participate in the catalytic cycle under these reaction conditions. In general, dienes and allenylic acetates should give the same products via similar alkylidene η^3-allyl palladium intermediates.[13,17] Accordingly, the main reason for the difference in reactivity between the dienes and allenylic acetates may be the challenging 1,3-oxidative addition with dienes (*Z*)-**6** and (*E*)-**6**.

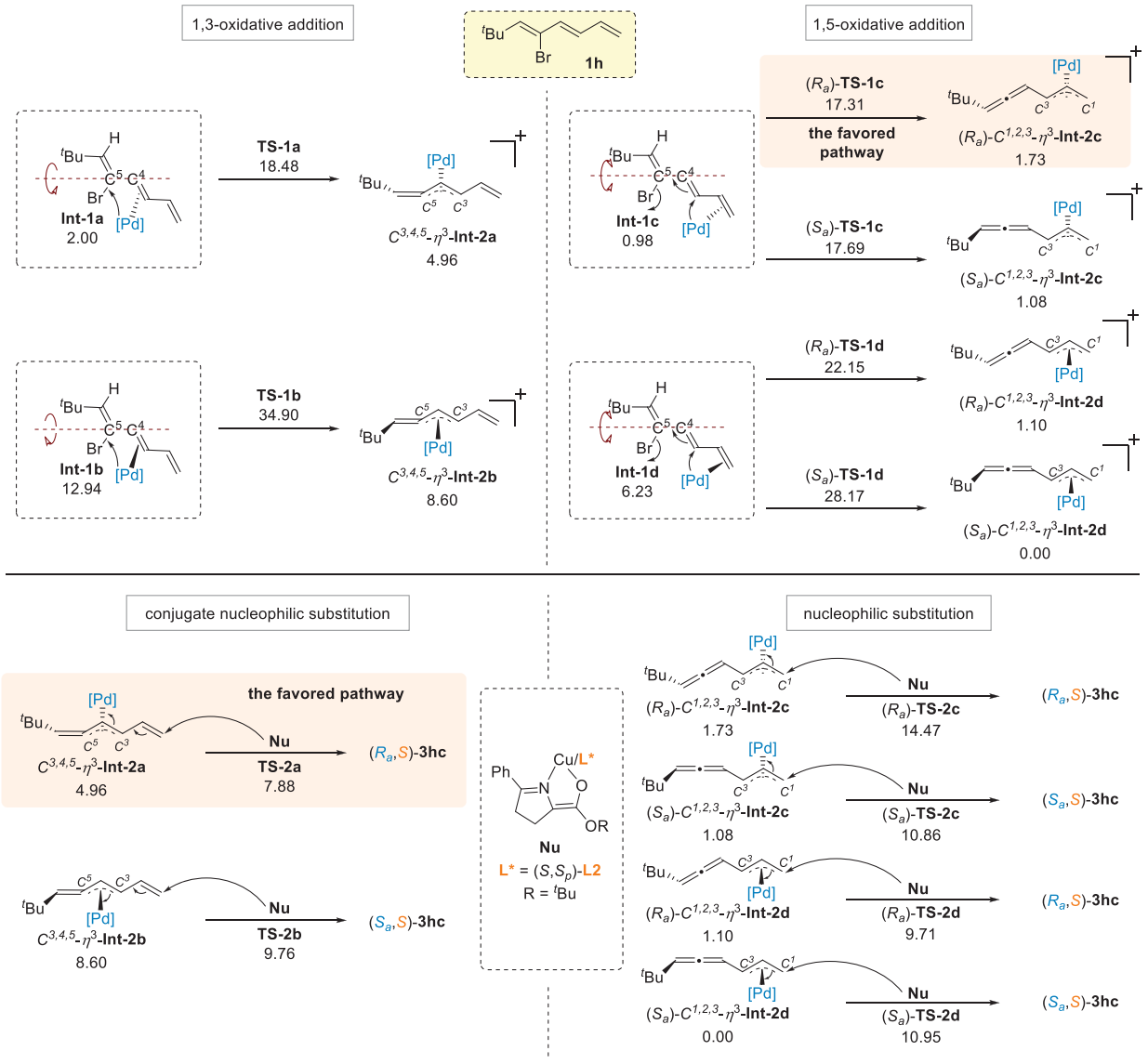
Subsequently, DFT calculations for both 1,3- and 1,5-oxidative addition processes with triene **1h** were conducted (Scheme 5, top). In general, all four possible η^2-complexes were initially formed when the Pd catalyst coordinated with the internal C^3=C^4 bond (**Int-1a** and **Int-1b**) and the terminal C^1=C^2 bond (**Int-1c** and **Int-1d**). For the 1,3-oxidative addition of **Int-1a** and **Int-1b**, the Br atom should approach the coordinated Pd catalyst to form a *cis*-conformation through C^4−C^5 σ-bond rotation, and then the coordinated Pd catalyst inserts into the C^5−Br bond to form two internal η^3-allyl palladium intermediates $C^{3,4,5}$-η^3-Int-**2a** and $C^{3,4,5}$-η^3-Int-**2b** via **TS-1a** (18.48 kcal/mol) and **TS-1b** (34.90 kcal/mol), respectively. For the 1,5-oxidation addition of **Int-1c** and **Int-1d**, the Br atom could move toward the same or opposite side of the coordinated Pd catalyst, and a S_N2'-type substitution occurs to afford four terminal η^3-allyl palladium intermediates [(R_a)-$C^{1,2,3}$-**Int-2c-2d** and (S_a)-$C^{1,2,3}$- **Int-2c-2d**] via (R_a)-**TS-1c** (17.31 kcal/mol), (S_a)-**TS-1c** (17.69 kcal/mol), (R_a)-**TS-1d** (22.15 kcal/mol), and (S_a)-**TS-1d** (28.17 kcal/mol), respectively. These DFT calculation results suggest that the 1,5-oxidative addition of **Int-1c** via (R_a)-**TS-1c** to afford (R_a)-$C^{1,2,3}$-**Int-2c** is favored with the lowest energy barrier ($\Delta G^{\ddagger}$ = 17.31 kcal/mol), which is in agreement with the control experiments.

Next, nucleophilic substitution of the electrophilic η^3-allyl palladium intermediates with nucleophiles was investigated by DFT calculations. Despite some of the η^3-allyl palladium intermediates being difficult to obtain through oxidative additions with **1h**, such as $C^{3,4,5}$-η^3-**Int-2a** and $C^{3,4,5}$-η^3-**Int-2b** (Scheme 5, top), these intermediates could be indirectly generated through classic η^3−η^1−η^3 interconversions from

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Scheme 5. DFT Calculation Results for All Six Possible Oxidative Additions and Nucleophilic Substitution Pathways for Triene 1h



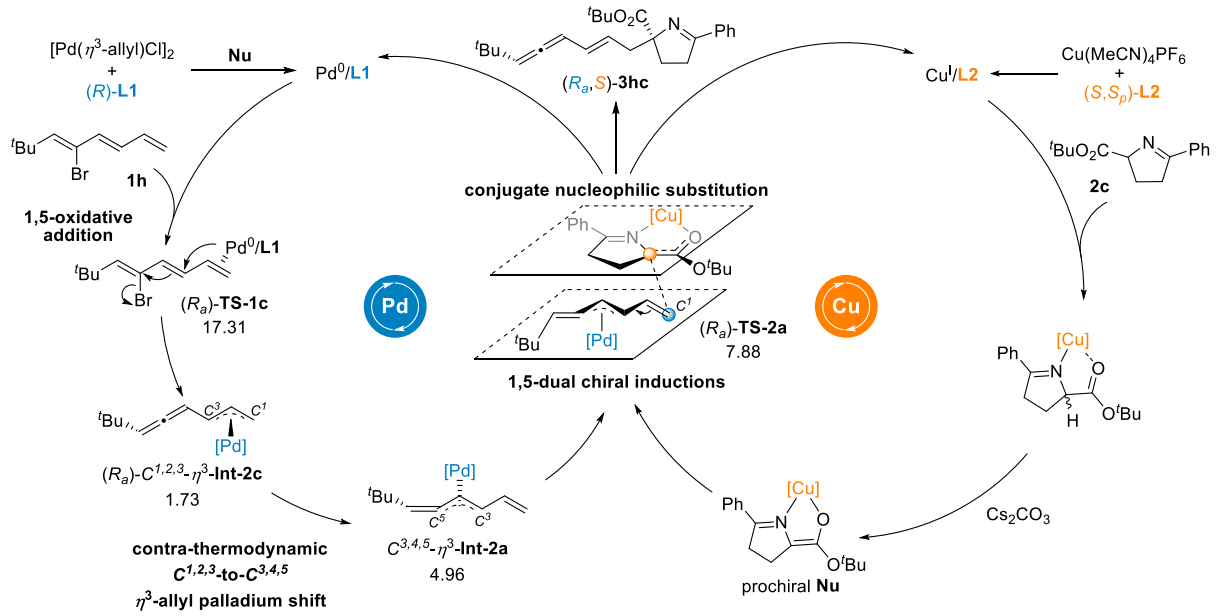
other η^3-allyl palladium intermediates. Thus, the corresponding nucleophilic substitution pathways involving all six η^3-allyl palladium intermediates [$C^{3,4,5}$-η^3-**Int-2a**,**2b** and (R_a)/(S_a)-$C^{1,2,3}$-η^3-**Int-2c**,**2d**] were investigated (Scheme 5, bottom). According to the Curtin−Hammett principle, the ratio of products is mainly controlled by the difference ($\Delta\Delta G^{\ddagger}$) in the free energies of the transition states of the nucleophilic substitution step, not depending on the free energies of the different intermediates. These results of the DFT calculations concerning the nucleophilic substitution process suggest that the conjugate nucleophilic substitution of $C^{3,4,5}$-η^3-**Int-2a** with **Nu** via (R_a)-**TS-2a** to afford the major product (R_a,S)-**3hc** is favored with the lowest energy barrier ($\Delta G^{\ddagger}$ = 7.88 kcal/mol). Based on the above studies on the oxidative addition and nucleophilic substitution, a mechanism is proposed as follows (Scheme 6): the Pd catalyst initially reacts with triene **1h**, affording (R_a)-$C^{1,2,3}$-η^3-**Int-2c** via a 1,5-oxidative addition [(R_a)-**TS-1c**]; then, (R_a)-$C^{1,2,3}$-η^3-**Int-2c** (1.73 kcal/mol) is converted into the less stable $C^{3,4,5}$-η^3-**Int-2a** (4.96 kcal/mol) via a contra-thermodynamic $C^{1,2,3}$-to-$C^{3,4,5}$ η^3-allyl palladium shift, while the Cu catalyst reacts with imino ester **2c** under the assistance of Cs_2CO_3, affording the nucleophilic azomethine ylides **Nu**; finally, $C^{3,4,5}$-η^3-**Int-2a** is attacked by the nucleophile via a conjugate nucleophilic substitution [(R_a)-**TS-2a**], giving the desired product (R_a,S)-**3hc** and regenerating both the Pd catalyst and Cu catalyst.

To further verify the η^3-allyl palladium shift, a control experiment with racemic allenyl acetate *rac*-**9** (1.0 equiv) was conducted with a [Pd/(R)-**L1** + Cu/(S,S_p)-**L3**] catalyst combination by using imino ester **2a** as the nucleophile precursor (Scheme 7). The desired product (R_a,S)-**3ha** was obtained with excellent stereocontrol of the axial chirality (99% ee and >20:1 dr), and the substrate **9** was recovered in 41% yield with 0% ee. Moreover, optically pure substrate (R_a)-**9** (>99% ee) was also investigated under the same reaction conditions. No racemization was observed (93% recovered yield, >99% ee), indicating that two enantiomers of the axially chiral allene substrate could not be converted into each other. The results suggest that the allenyl axial chirality should be racemized via the contra-thermodynamic $C^{1,2,3}$-to-$C^{3,4,5}$ η^3-allyl

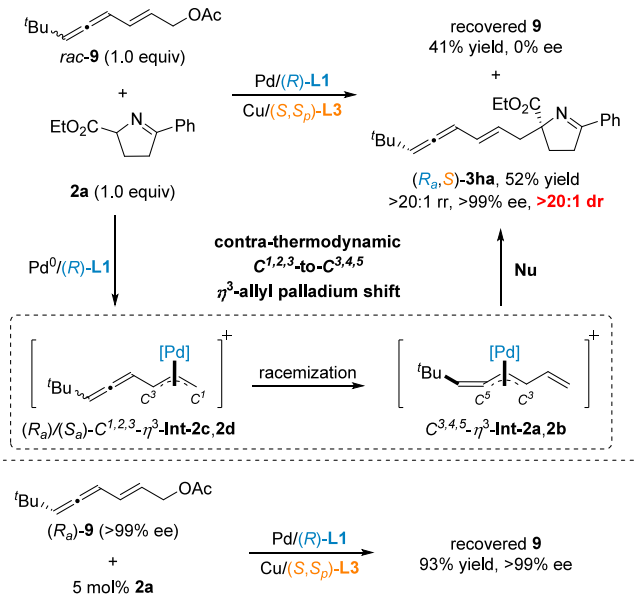
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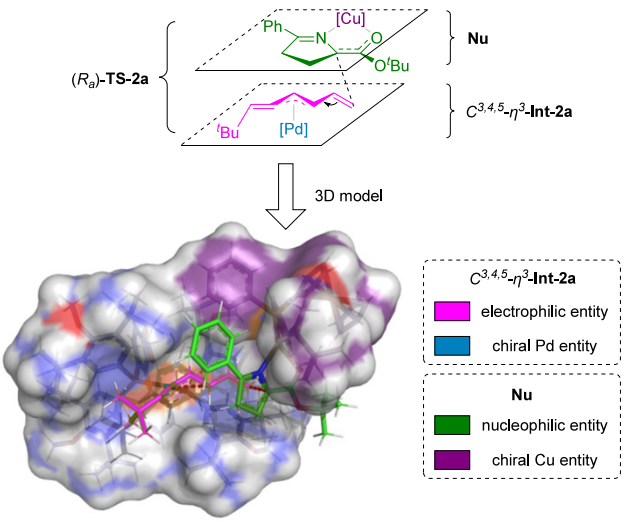
Scheme 6. Proposed Mechanism for the 1,5-Double Chiral Induction Model



Scheme 7. Control Experiments with Racemic Allylic Acetate *rac*-9 and Enantiopure Allylic Acetate (R_a)-9



Scheme 8. 3D Diagram of (R_a)-TS-2a Representing the 1,5-Double Chiral Induction Model



palladium shift. This unusual elementary reaction could provide an efficient strategy for remote dynamic kinetic asymmetric transformations of racemic allenes.

Finally, the 1,5-double chiral induction model in the conjugate nucleophilic substitution is well represented by the 3D diagram of (R_a)-**TS-2a** (Scheme 8). The elaborate chiral environment is established by the synergistic bimetallic catalytic system consisting of the Pd catalyst (blue) and the Cu catalyst (purple) for both the electrophilic entity (pink) and the nucleophilic entity (green). Both the allenyl axial chirality and remote central chirality are concurrently constructed with excellent stereoselectivity, benefiting from the superiority of the enzyme-like pocket established by the bimetallic catalytic system.

All of the above mechanistic studies suggest that the C═C bond relay handle plays an important role in this reaction: (I) The terminal C═C bond provides an appropriate coordination site for the Pd catalyst; therefore, a 1,5-oxidative addition occurs smoothly with a lower energy barrier to afford the terminal η^3-allyl palladium intermediate (R_a)-$C^{1,2,3}$-η^3-**Int-2c**; (II) subsequently, a contra-thermodynamic η^3-allyl palladium shift occurs along the unsaturated C═C bond, forming an internal η^3-allyl palladium intermediate $C^{3,4,5}$-η^3-**Int-2a**, which is beneficial for the following nucleophilic substitution process; and (III) the conjugate nucleophilic substitution proceeds smoothly through the attack of the nucleophile **Nu** at the terminal C═C bond, which avoids the crowded three-dimensional environment of two chiral metal intermediates and guarantees the efficient asymmetric inductions of allenyl axial chirality and remote central chirality.

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CONCLUSIONS

We have developed a protocol for 1,5-double chiral induction based on a C=C bond relay strategy via a synergistic Pd/Cu catalytic system, which was successfully applied to the asymmetric construction of 1,5-nonadjacent stereocenters bearing allenyl axial chirality and central chirality in a single operation. A wide range of axial chiral allenes bearing a remote stereocenter were readily synthesized in an enantio- and diastereodivergent manner under mild reaction conditions. Based on the control experiments and DFT calculations, a reasonable mechanism with three key elementary reactions for this 1,5-double chiral induction model involving 1,5-oxidative addition, a contra-thermodynamic η^3-allyl palladium shift, and conjugate nucleophilic substitution has been proposed, demonstrating the importance of the C=C bond relay strategy for the reactivity and enantio- and diastereoselectivity of the reaction. This strategy may provide a general avenue for the asymmetric synthesis of chiral molecules bearing remote nonadjacent stereocenters.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.4c00497.

Experimental procedures, computational details, and characterization data for all reactions and products, including ^{1}H and ^{13}C NMR spectra, ^{19}F NMR spectra, HPLC spectra, HRMS, and crystallographic data for (R_a,S)-**3hi** (PDF)

Accession Codes

CCDC 2216926 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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